Energy Transformations — Practice

Chapter 6 · Energy Transformations + Engineering Design · Week 7

Name: _____ Date: _____

A. TRANSFER / TRANSFORMATION / BOTH

Read each system. Circle TRANSFER, TRANSFORMATION, or BOTH.

1. Warm water warms a cool spoon.
   TRANSFER **TRANSFORMATION** **BOTH**

2. A flashlight changes stored energy to light.
   TRANSFER **TRANSFORMATION** **BOTH**

3. A toy car hits a cup: the cup moves and we hear a sound.
   TRANSFER **TRANSFORMATION** **BOTH**

4. A buzzer changes electrical energy to sound.
   TRANSFER **TRANSFORMATION** **BOTH**

5. Falling dominoes knock over the next domino, and we hear a click.
   TRANSFER **TRANSFORMATION** **BOTH**

B. INPUT → CHANGE → OUTPUT

Fill in the missing part of each energy chain.

1. Wind-up toy: INPUT = stored energy → CHANGE = _____ → OUTPUT = the toy moves.

2. Flashlight: INPUT = stored energy → CHANGE = _____ → OUTPUT = light we see.

3. Solar panel: INPUT = _____ → CHANGE = light to electrical → OUTPUT = electricity.

C. FIX THE ENERGY CHAIN

Each chain has a mistake. Rewrite it correctly.

1. *"INPUT: sound → OUTPUT: stored energy" (the order is wrong)*

2. *"The energy DISAPPEARED when the toy stopped."*

D. DESIGN THINKING — criteria vs constraints

Label each statement CRITERIA (what it must do) or CONSTRAINT (a limit).

1. It must make a clear sound signal. → _____
2. You may use only the provided materials. → _____
3. It must work three times in a row. → _____
4. No wall electricity is allowed. → _____

E. SHORT EXPLANATION

How can changing energy from one form to another solve a problem? Use one device as an example (toaster, flashlight, wind-up timer, or solar panel).